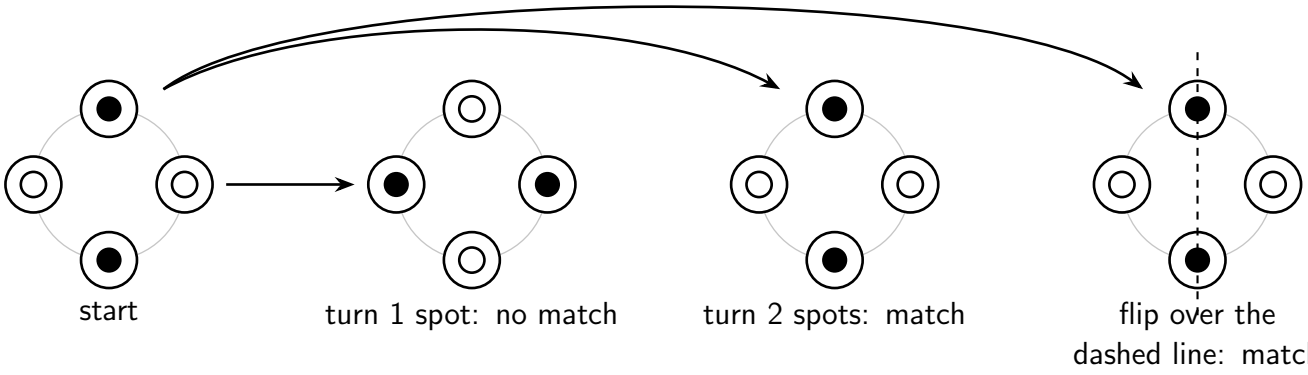
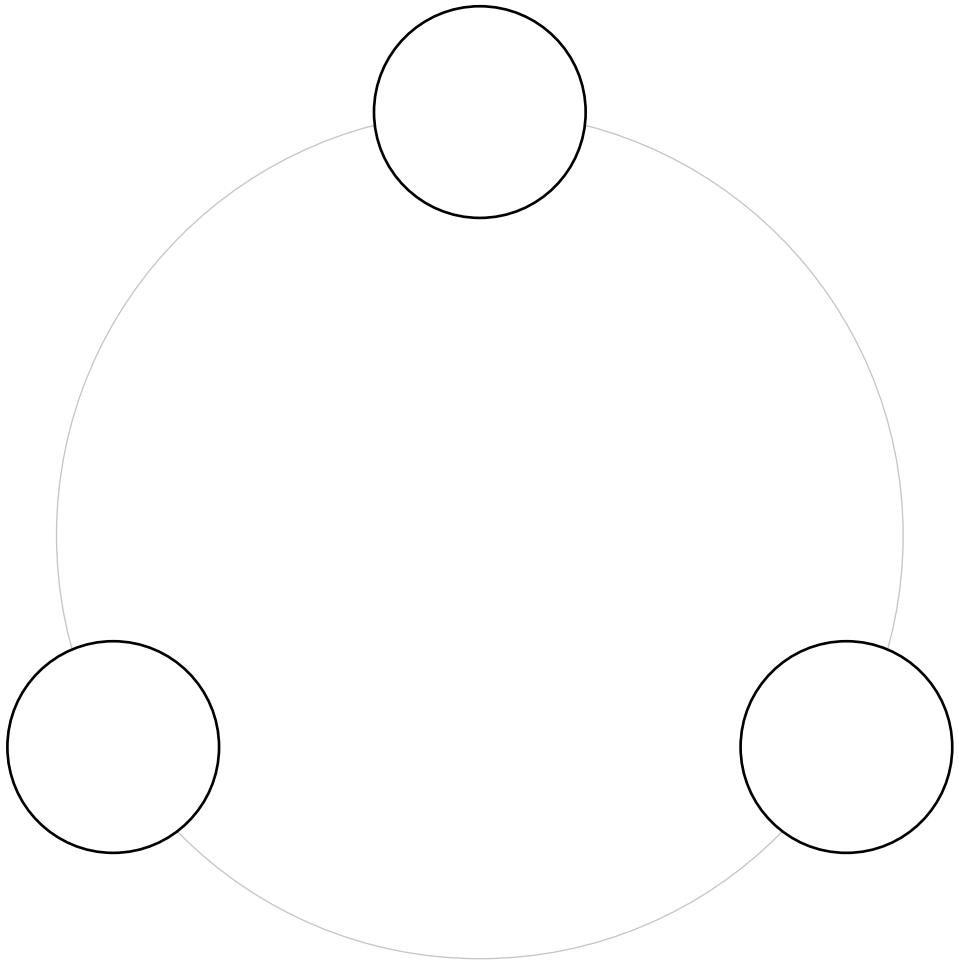
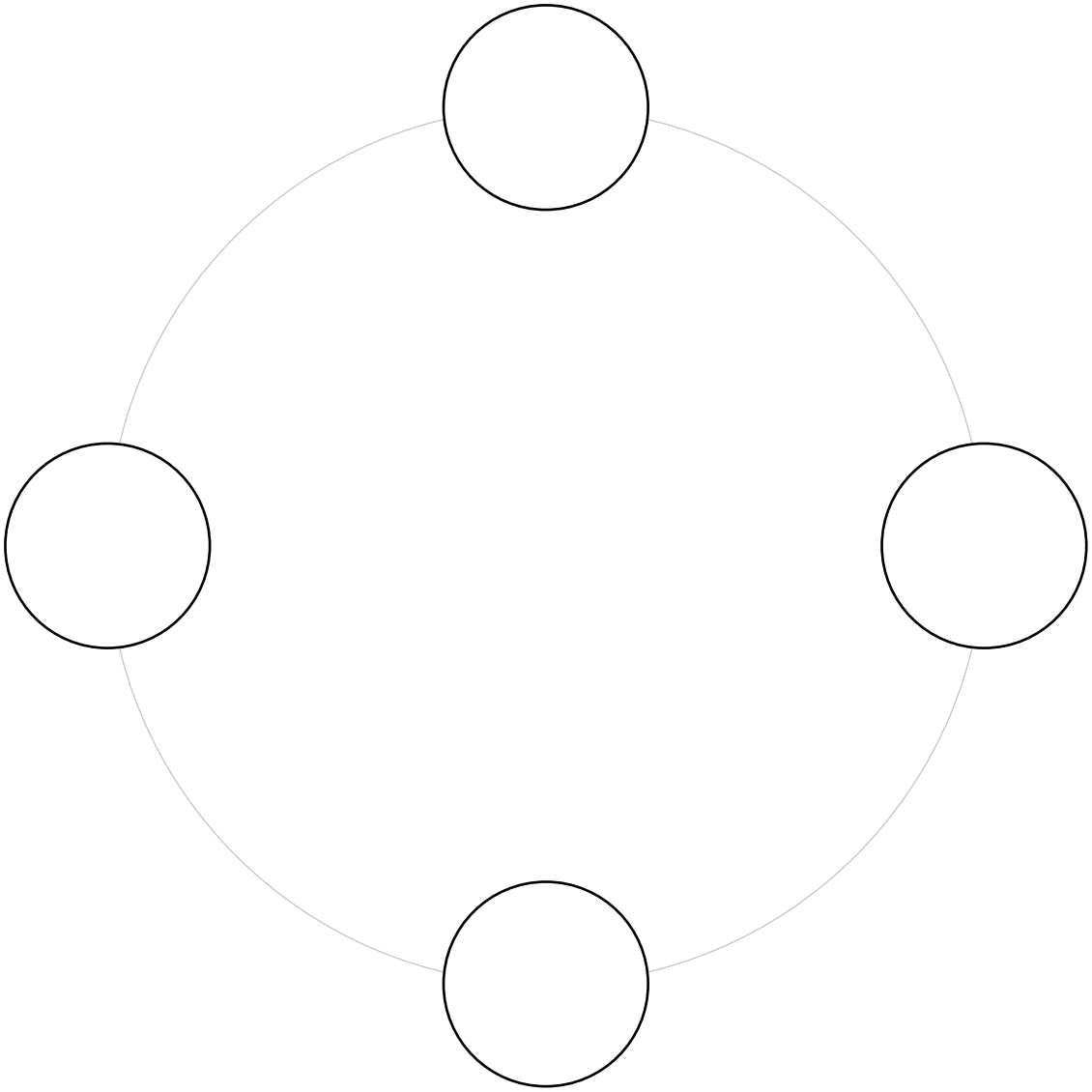
Fill every spot; neighbors may be the same kind. Only counter kinds and spots count. Copy a pattern on tracing paper with a dot of each counter's color. One partner builds. The other keeps the original still and turns or flips (turns over) the copy to make every dot sit on its own color. A whole turn does not count.



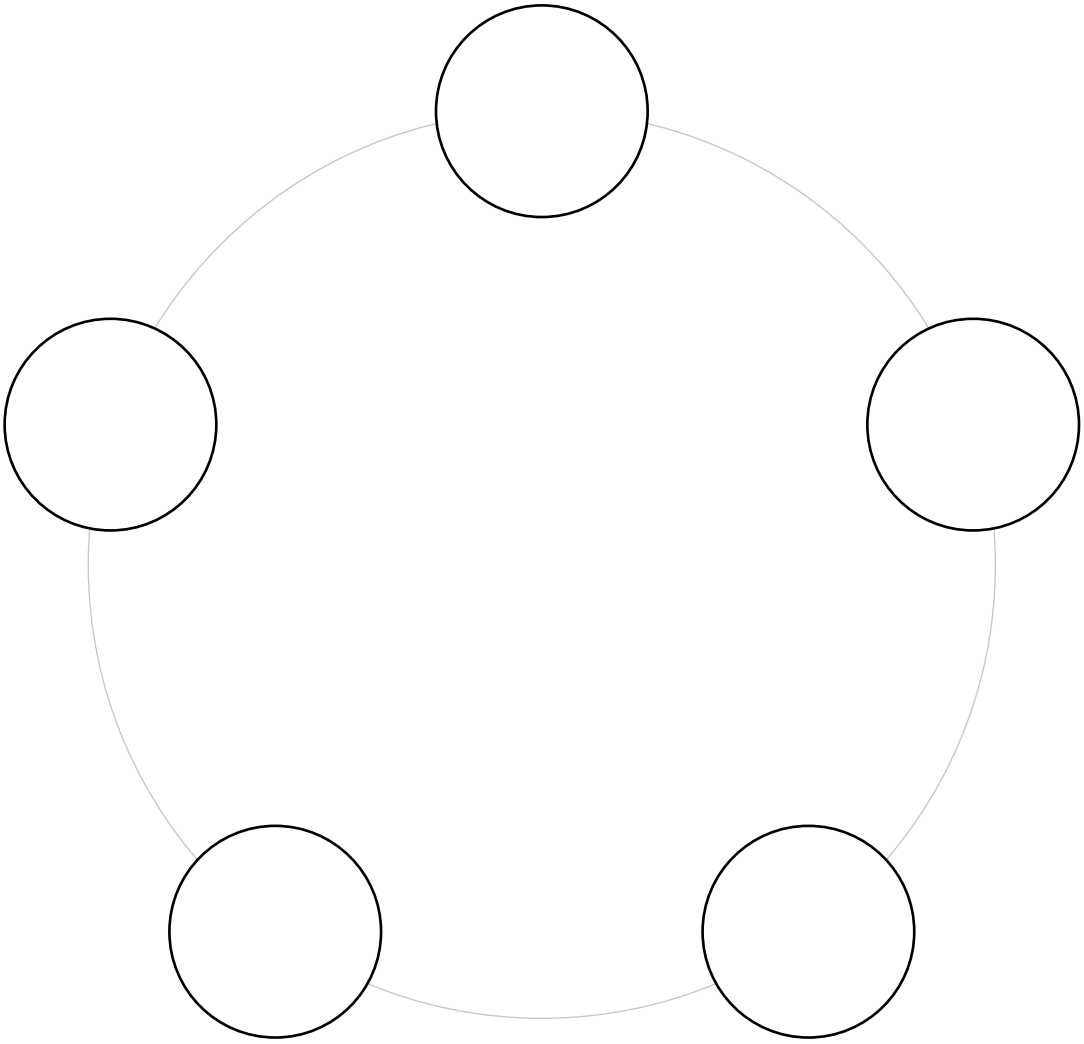
Problem 1: Find the fewest kinds of counter for a pattern on the three-ring that no turn or flip can match. Can you rule out every pattern with fewer kinds?



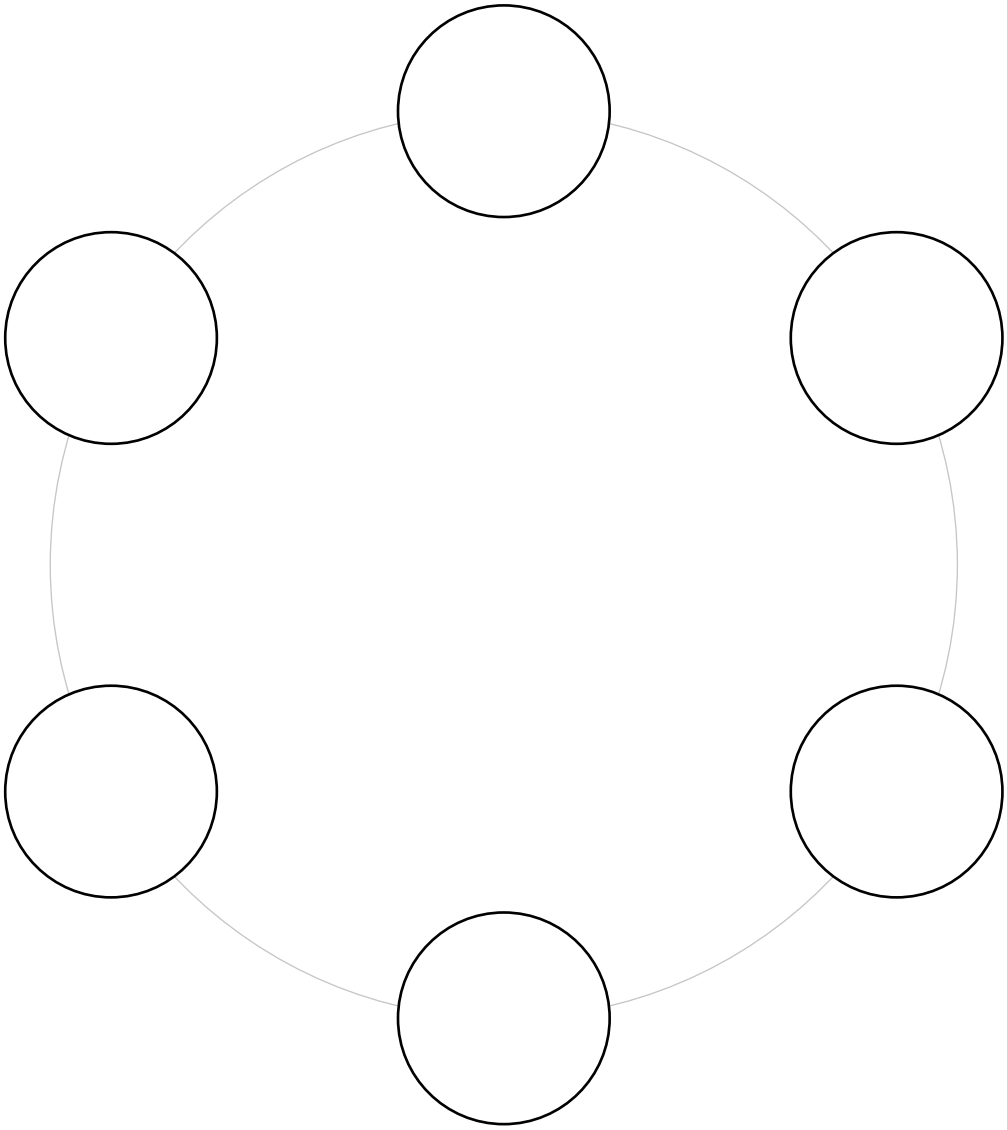
Problem 2: Find the fewest kinds of counter for a pattern on the four-ring that no turn or flip can match. Explain why fewer kinds cannot work.



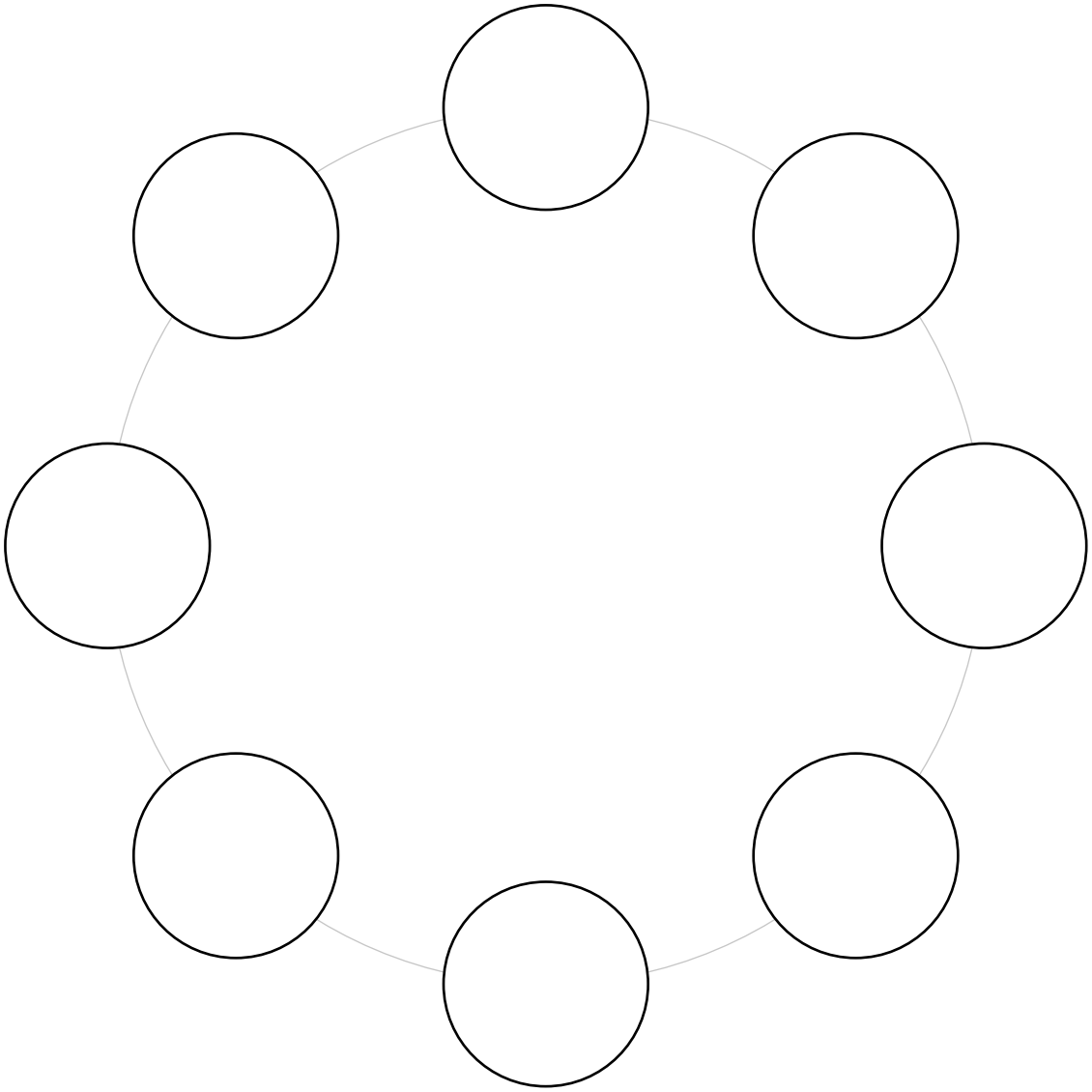
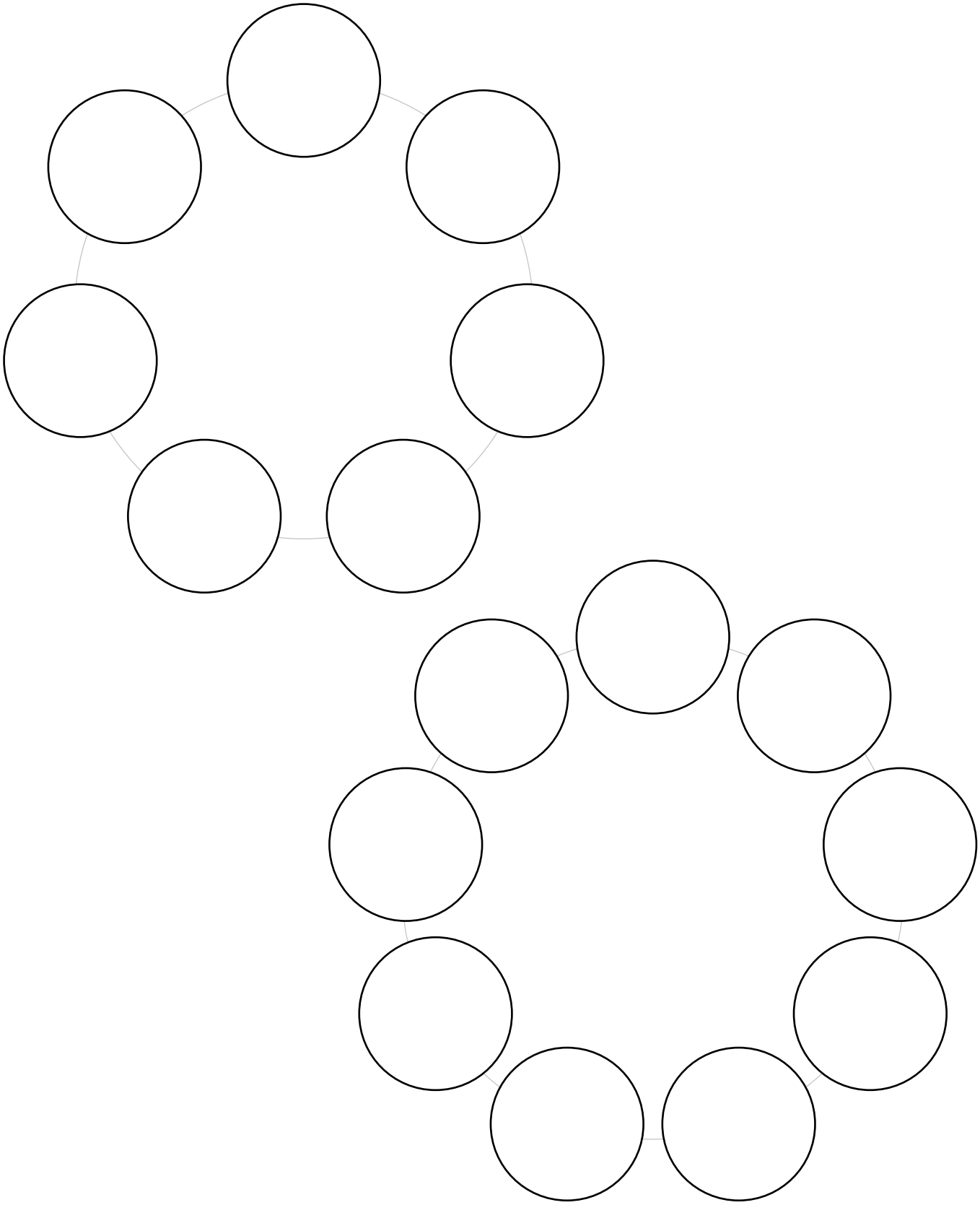
Problem 3: Find the fewest kinds of counter for a pattern on the five-ring that no turn or flip can match. Find an explanation that also works for the three-ring and four-ring.



Problem 4: Use two kinds of counter to make a pattern on this six-ring that no turn or flip can match. Explain why no turn or flip can match it.



Problem 5: Can you make a two-kind pattern that no turn or flip can match on any ring with at least six spots? Test your idea on this eight-ring, then explain why it always works.



Extra workspace for Problem 5: a seven-ring and a nine-ring.